Final Examination, Session 2, 2004
Model Answers

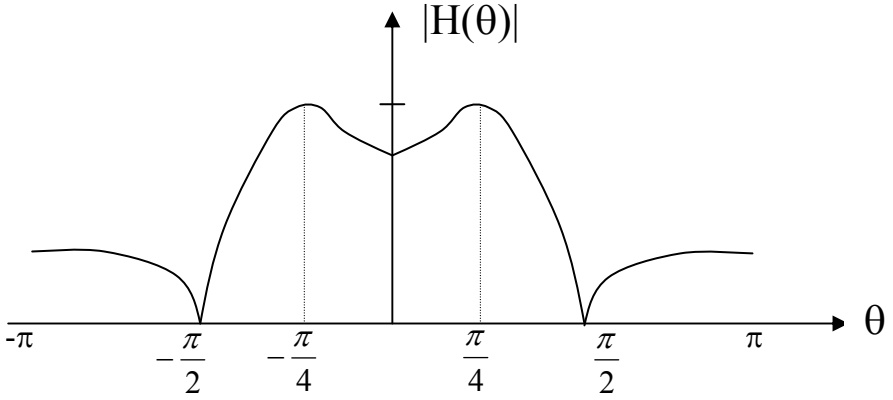
Question 1

(a)

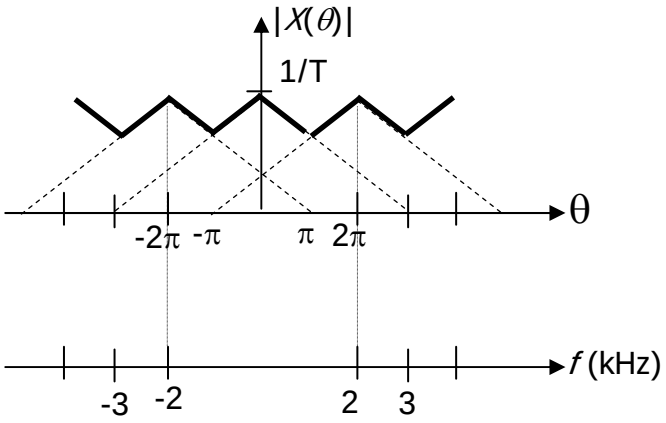
$$H(z) = K\ \frac{(z + j)(z - j)}{(z - 0.5 - 0.5j)(z - 0.5 + 0.5j)}$$

$$= K\ \frac{(1 + z^{-2})}{1 - z^{-1} + 0.5z^{-2}}$$

(ii)



(b)



(c)

$$H(z) = \frac{Y(z)}{X(z)} = \frac{b}{1 - az^{-1}}$$

$$\left|H(\theta)\right|_{\theta = 0} = 1$$

$$\therefore\ 1 = \frac{b}{1 - a}$$

$$\therefore b = 1 - a$$

Normalised filter:

$$H(z) = \frac{1 - a}{1 - az^{-1}}$$

$$|H(\theta)|^2 = \left| \frac{(1 - a)^2}{(1 - a\cos\theta)^2 + a^2\sin^2\theta} \right|$$

$$= \frac{(1 - a)}{1 - a^2 - 2a\cos\theta}$$

Let 3 – dB cut off be θc

$$\therefore \frac{1}{2}|H(0)|^2 = |H(\theta_C)|^2$$

$$\frac{1}{2}.1 = \frac{(1 - a)^2}{1 - a^2 - 2a\cos\theta_c}$$

$$\therefore 1 + a^2 - 2(1 - a)^2 = 2a\cos\theta_c$$

$$\therefore \cos\theta_c = \frac{-a^2 + 4a - 1}{2a}$$

$$\theta_c = \cos^{-1}\left[\frac{-a^2 + 4a - 1}{2a}\right]$$

(d)

$$H_0 = 3 \qquad H_1 = 3(2 + j\sqrt{3})$$

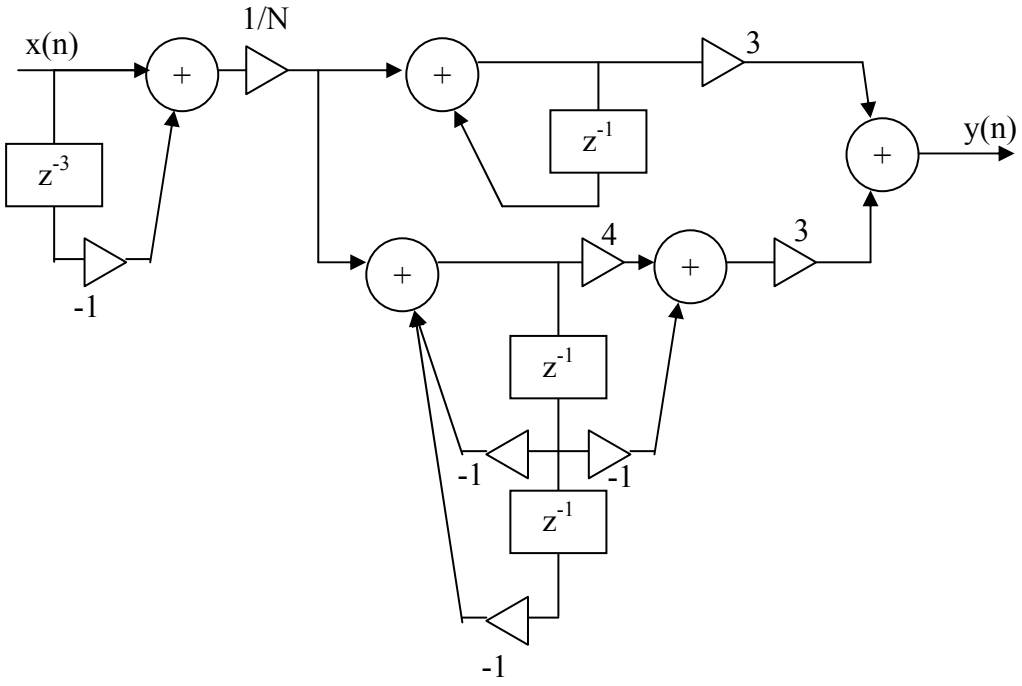
$$H_{-1} = H_1^* = 3(1 - j\sqrt{3})$$

$$N = 3$$

$$P(z) = \frac{1 - z^{-3}}{3}\left[\frac{3}{1 - z^{-1}} + \frac{3(2 + j\sqrt{3})}{1 - z^{-1}e^{\frac{j2\pi}{3}}} + \frac{3(2 - j\sqrt{3})}{1 - z^{-1}e^{\frac{j2\pi}{3}}}\right]$$

$$= \frac{1 - z^{-3}}{3}\left[\frac{3}{1 - z^{-1}} + \frac{3(2 + j\sqrt{3})(1 - z^{-1}e^{\frac{-j2\pi}{3}}) + (2 - j\sqrt{3})(1 - z^{-1}e^{\frac{j2\pi}{3}})}{1 + z^{-1} + z^{-2}}\right]$$

$$= \frac{1 - z^{-3}}{3}\left[\frac{3}{1 - z^{-1}} + \frac{3(4 - z^{-1})}{1 + z^{-1} + z^{-2}}\right]$$



(ii)

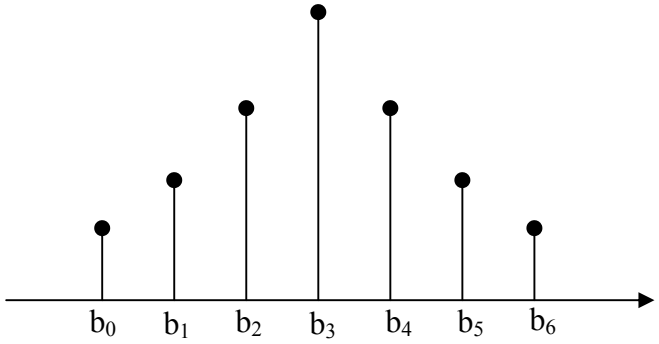
$$P(z) = \frac{1 - z^{-3}}{3}\left[\frac{3}{1 - z^{-1}} + \frac{3(4 - z^{-1})}{1 + z^{-1} + z^{-2}}\right]$$

$$= \frac{(1 - z^{-3})}{3} \cdot 3\left[\frac{5 - 4z^{-1} + 2z^{-2}}{(1 - z^{-1})(1 + z^{-1} + z^{-2})}\right]$$

$$P(z) = 5 - 4z^{-1} + 2z^{-2}$$

FIR filter: $h(0) = 5, h(1) = -4, h(2) = 2$

(E)



For a linear phase the coefficients must be symmetrical about b_3.
Therefore $b_4 = b_2 = 3$, $b_5 = b_1 = 2$. $b_6 = b_0 = 1$

(ii)

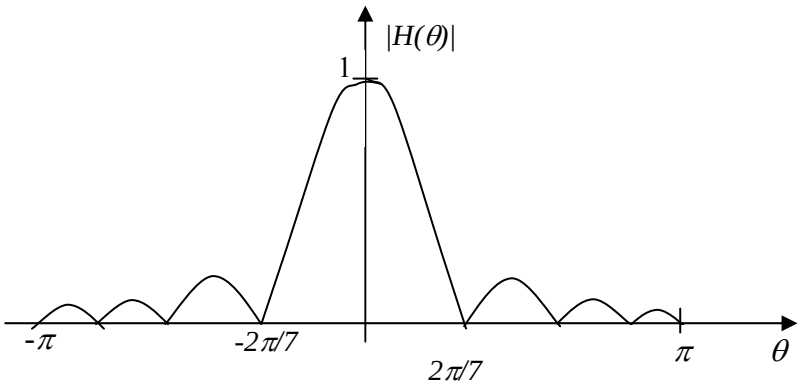
$$H(z) = 1 + z^{-1} + z^{-2} + z^{-3} + z^{-4} + z^{-5} + z^{-6}$$

$$= \frac{1 - z^{-N}}{1 - z^{-1}} = \frac{1 - z^{-7}}{1 - z^{-1}} \qquad N = 7$$

$$H(\theta) = \frac{1 - e^{-j7\theta}}{1 - e^{-j\theta}} = \frac{e^{\frac{-j7\theta}{2}}}{e^{\frac{-j\theta}{2}}}\left[\frac{e^{\frac{j7\theta}{2}} - e^{\frac{-j\theta}{2}}}{e^{\frac{j\theta}{2}} - e^{\frac{-j\theta}{2}}}\right]$$

$$H(\theta) = e^{-j3\theta}\,\frac{\sin\frac{7\theta}{2}}{\sin\frac{\theta}{2}} \qquad \therefore |H(\theta)| = \left|\frac{\sin\frac{7\theta}{2}}{\sin\frac{\theta}{2}}\right|$$

$$\therefore \phi(0) = -3\theta$$



(iii)

$$\therefore \phi(0) = -3\theta$$

$$\therefore \ \therefore \frac{-d\phi(\theta)}{d\theta} = 3 \longleftarrow \text{Constant}$$

Jump Delay

(F)

H(z) is an IIR filter.

(ii)

$$H(z) = \frac{\frac{1}{2}(1 - az^{-1}) + 1}{(1 - \frac{1}{2}z^{-1})(1 - az^{-1})} = \frac{\frac{1}{2} - \frac{a}{2}z^{-1} + 1}{1 - az^{-1} - \frac{1}{2}z^{-1} + \frac{a}{2}z^{-2}}$$

$$= \frac{\frac{3}{2} - \frac{a}{2}z^{-1}}{\frac{2 - (2a+1)z^{-1} + az^{-2}}{2}} = \frac{3 - az^{-1}}{2 - (2a+1)z^{-1} + az^{-2}}$$

$$2y(n) - (2a+1)y(n-1) + ay(n-2) = 3x(n) - ax(n-1)$$

$$\therefore y(n) = \frac{3}{2}x(n) - \frac{a}{2}x(n-1) + \frac{2a+1}{2}y(n-1) - \frac{a}{2}y(n-2)$$

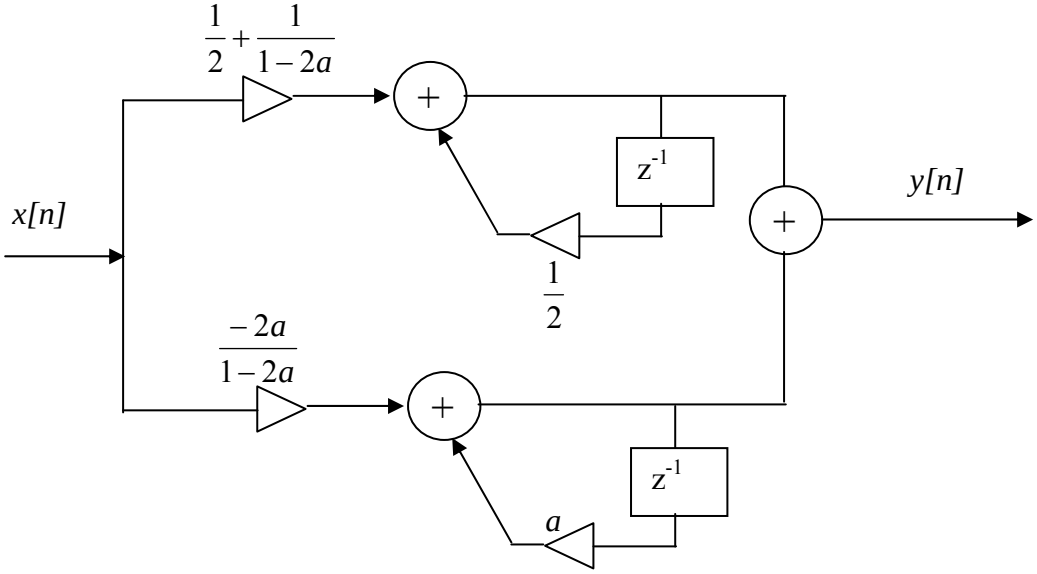
(iii)

$$H(z) = \frac{\frac{1}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{A}{1 - \frac{1}{2}z^{-1}} + \frac{B}{1 - az^{-1}}$$

$$\frac{1}{(1 - \frac{1}{2}z^{-1})(1 - az^{-1})} = \frac{\frac{1}{1-2a}}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{-2a}{1-2a}}{1 - az^{-1}} = \frac{\frac{1}{1-2a}}{1 - \frac{1}{2}z^{-1}} - \frac{\frac{2a}{1-2a}}{1 - az^{-1}}$$

$$H(z) = \frac{\frac{1}{2}}{1 - \frac{1}{2}z^{-1}} + \frac{\frac{1}{1-2a}}{1 - \frac{1}{2}z^{-1}} - \frac{\frac{2a}{1-2a}}{1 - az^{-1}}$$

$$H(z) = \frac{(\frac{1}{2} + \frac{1}{1-2a})}{1 - \frac{1}{2}z^{-1}} + \frac{-(\frac{2a}{1-2a})}{1 - az^{-1}}$$



(iv)

$$h(n) = (\frac{1}{2} + \frac{1}{1-2a})(\frac{1}{2})^n u(n) - (\frac{2a}{1-2a})a^n u(n)$$

When $a < 1$, h(n) will decrease as n increases.

If $a > 1$, h(n) will increase as n increases.

(g)

$$H(z) = \frac{3 + 3z^{-1}(e^{-2T} - 2e^{-T})}{(1 - e^{-2T}z^{-1})(1 - e^{-T}z^{-1})}$$

$$= \frac{A}{1 - e^{-2T}z^{-1}} + \frac{B}{1 - e^{-T}z^{-1}}$$

$$H(z) = \frac{6}{1 - e^{-2T}z^{-1}} - \frac{3}{1 - e^{-T}z^{-1}} \qquad \text{NOTE: } \frac{1}{s+a} \rightarrow \frac{1}{1 - e^{-aT}z^{-1}}$$

$$\therefore H(s) = \frac{6}{s+2} - \frac{3}{s+1} = \frac{3s}{(s+1)(s+2)}$$

Question 2.

$$H(z) = \frac{z}{z^2 - 2\cos\theta_0 z + 1} = \frac{z}{z^2 - b_1 z + 1}$$

From z transform table

$$\text{Sin } n\theta_0 \; u(n) \leftrightarrow \frac{z\text{Sin}\theta_0}{z^2 - 2\cos\theta_0 z + 1}$$

$$\therefore \frac{\text{Sin}(n\theta_0)}{\text{Sin}\theta_0} u(n) \leftrightarrow \frac{z}{z^2 - 2\cos\theta_0 z + 1}$$

H(z)

$$\therefore h(n) = \frac{\sin n_0}{\sin\theta} u(n)$$

(ii)

$$b_1 = 2\cos\theta_0$$

$$\Delta b_1 = -2\sin\theta_0 . \Delta\theta_0$$

$$\Delta b_1 = -2\sin\theta_0 . 2\pi \frac{\Delta f_0}{f_s}$$

$$\Delta f_0 = \frac{-f_s \Delta b_1}{4\pi \sin\left(\frac{2\pi f_0}{f_s}\right)}$$

(iii)

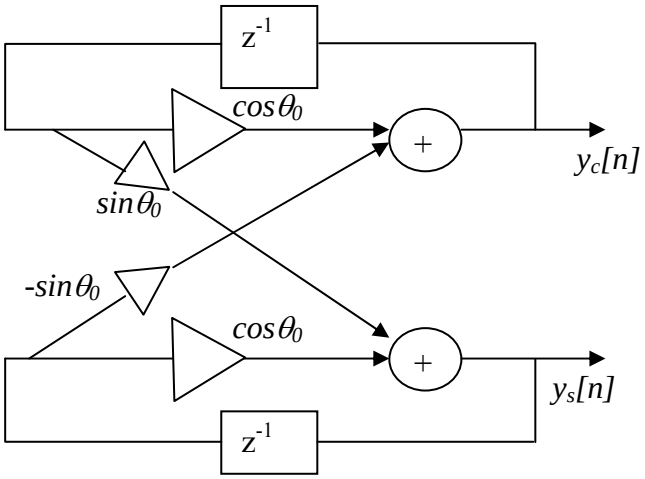
Δf_0 (min) is high resolution when $\sin(2\pi \frac{f_0}{f_s}) = 1$

$$\therefore |\Delta f_0| = \frac{f_s \Delta b_1}{4\pi}$$

(b)

$$y_c(n) = \cos\theta_0\ y_c(n-1) - \sin\theta_0\ y_s(n-1)$$

$$y_s(n) = \sin\theta_0\ y_c(n-1) + \cos\theta_0\ y_s(n-1)$$



n = 0 $\quad y_s(0) = \sin\theta_0\ (A\cos\theta_0) + \cos\theta_0(\sin\theta_0) = 0$

$$y_c(0) = \cos\theta_0\ (A\cos\theta_0) - \sin\theta_0(-\sin\theta_0) = 1$$

$$y(0) = 1$$

n = 1 $\quad y_s(1) = \sin\theta_0 + 0 = \sin\theta_0$

$$y_c(1) = \cos\theta_0 - \sin\theta_0.0 = \cos\theta_0$$

$$y_c(1) = \cos\theta_0$$

n = 2 $\quad y_c(2) = \cos\theta_0 y_c(1) - \sin\theta_0 y_s(1)$

$$y_c(2) = \cos 2\theta_0$$

$$y_c(n) = \cos n\theta_0$$

(c)

$$H(z) = \frac{2z^3 - 4z^2 + 9}{(z-3)(z^2 + z + 0.5)}$$

Since H(z) has a pole at z = 3, the given transfer function is unstable.

To construct a stable transfer function having the same magnitude response consider another transfer function G(z).

$$G(z) = \frac{2z^3 - 4z^2 + 9}{(1-3z)(z^2 + z + 0.5)} = H(z)A(z)$$

where $A(z) = \dfrac{1-3z}{z-3}$ is an all pass function.

Thus $|G(\theta)| = |H(\theta)||A(\theta)|$

$$\| \\ \mathbf{1}$$

$$\therefore |G(\theta)| = |H(\theta)|$$

Question 3

The monotonic requirements indicate a Butterworth filter.

$$\theta_c = 2\pi \frac{f_c}{f_s} = 2\pi \frac{10{,}000}{100{,}000} = 0.2\pi$$

$$\theta_h = 2\pi \frac{f_h}{f_s} = 2\pi \frac{20{,}000}{100{,}000} = 0.4\pi$$

Prewarping

$$w = \frac{2}{T} \tan \frac{\theta}{2}$$

$$w_c' = \frac{2}{T}\tan\frac{0.2\pi}{2} = 2f_s \tan(0.1\pi) = 0.6498\times10^5$$

$$w_h' = 2f_s \tan\left(\frac{0.4\pi}{2}\right) = 1.4531\times10^5$$

Butterworth Filter Design

$$|H_a(w)|^2 = \frac{1}{1+\left(\frac{w_h'}{w_c'}\right)^{2N}}$$

To ensure 10dB attenuation at w_h'

$$10\log|H_a(w)|^2 = -10\log\left[1+\left(\frac{1.4531}{0.6498}\right)^{2N}\right]$$

$$\begin{matrix}10\log|H_a(w)|^2 \\ \| \\ -10\text{ dB}\end{matrix} = -10\log\left[1+\left(\frac{1.4531}{0.6498}\right)^{2N}\right]$$

$$10 = 1 + \left[\frac{1.4531}{0.6498}\right]^{2N}$$

$$\therefore n = 1.367$$

Select n = 2

$$H(s) = \frac{1}{s^2+\sqrt{2}s+1}$$

$$H(s) = \frac{(w_c')^2}{s^2+\sqrt{2}w_c'+(w_c')^2} \qquad s \to \frac{2}{T}.\frac{1-z^{-1}}{1+z^{-1}}$$

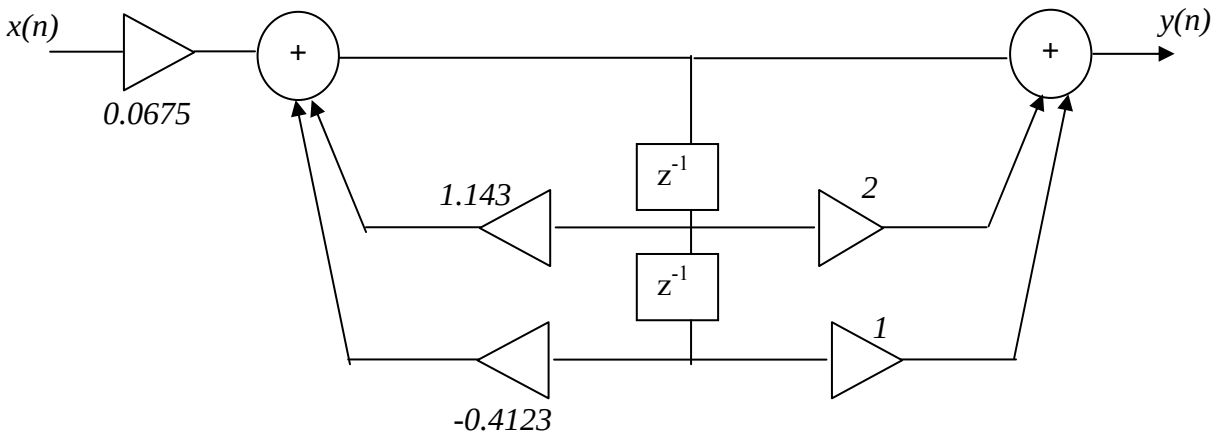
$$H(z) = \frac{\left(\frac{2}{T}\right)^2(\tan(0.1\pi))^2}{\left(\frac{2}{T}\right)^2\left(\frac{1-z^{-1}}{1+z^{-1}}\right)^2+\sqrt{2}\left(\frac{2}{T}\right)\left(\frac{1-z^{-1}}{1+z^{-1}}\right)\tan(0.1\pi)+\left(\frac{2}{T}\right)^2\tan(0.1\pi)^2+\left(\frac{2}{T}\right)\left(\frac{1-z^{-1}}{1+z^{-1}}\right)}$$

$$= \frac{[\tan(0.1\pi)]^2 (1+z^{-1})^2}{(1-z^{-1})^2 + \sqrt{2}\tan(0.1\pi)(1-z^{-2}) + [\tan(0.1\pi)]^2 (1+z^{-1})^2}$$

$$= \frac{0.1056(1+2z^{-1}+z^{-2})}{1-2z^{-1}+z^{-2}+0.46-0.46z^{-2}+0.1056+0.211z^{-1}+0.1056z^{-2}}$$

$$= \frac{0.1056(1+2z^{-1}+z^{-2})}{1.5656-1.789z^{-1}+0.6456z^{-2}}$$

$$= \frac{0.0675(1+2z^{-1}+z^{-2})}{1-1.143z^{-1}+0.4123z^{-2}}$$



(B)

N = 7 (given)

$$\phi(\theta) = -\beta\theta = -\left(\frac{n-1}{2}\right)\theta = -\left(\frac{7-1}{2}\right)\theta = -3\theta$$

$$\therefore \beta = 3$$

(ii)

$$h_d(n) = \frac{1}{2\pi}\int_{\frac{-\pi}{7}}^{\frac{\pi}{7}} e^{-j3\theta} e^{jn\theta}\, d\theta$$

$$= \frac{1}{2\pi j(n-3)}\left[e^{j(n-3)\frac{\pi}{7}} - e^{-j(n-3)\frac{\pi}{7}} \right] \qquad n \neq 3$$

$$= \frac{\sin (n-3)\frac{\pi}{7}}{\pi (n-3)} \qquad n \neq 3$$

$$h_d(3) = \frac{1}{7}$$

(iii)

$$h(n) = h_d(n) W_B(n)$$

$$\left.\begin{aligned} W_B(0) &= 0 \\ W_B(1) &= \frac{1}{3} \\ W_B(2) &= \frac{2}{3} \\ W_B(3) &= 1 \\ W_B(4) &= \frac{2}{3} \\ W_B(5) &= \frac{1}{3} \\ W_B(6) &= 0 \end{aligned}\right\} \text{ Bartlett Window}$$

$$h(0) = h_d(0) W_B(0) = 0$$

$$h(1) = h(5)$$

$$h(2) = h(4)$$

$$h(3) = h_d(3) W_B(3) = \frac{1}{7}$$

$$h(4) = h_d(4) W_B(4) = 0.921$$

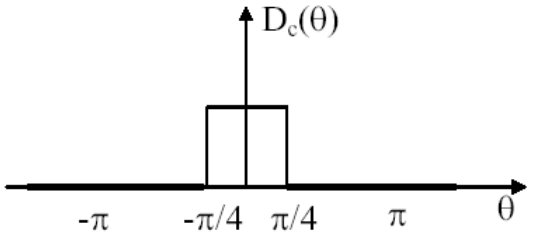
$$h(5) = 0.041$$

$$h(6) = 0$$

Question 4

(A)

$B_c(\theta) = 1, D_c(\theta) = 1$

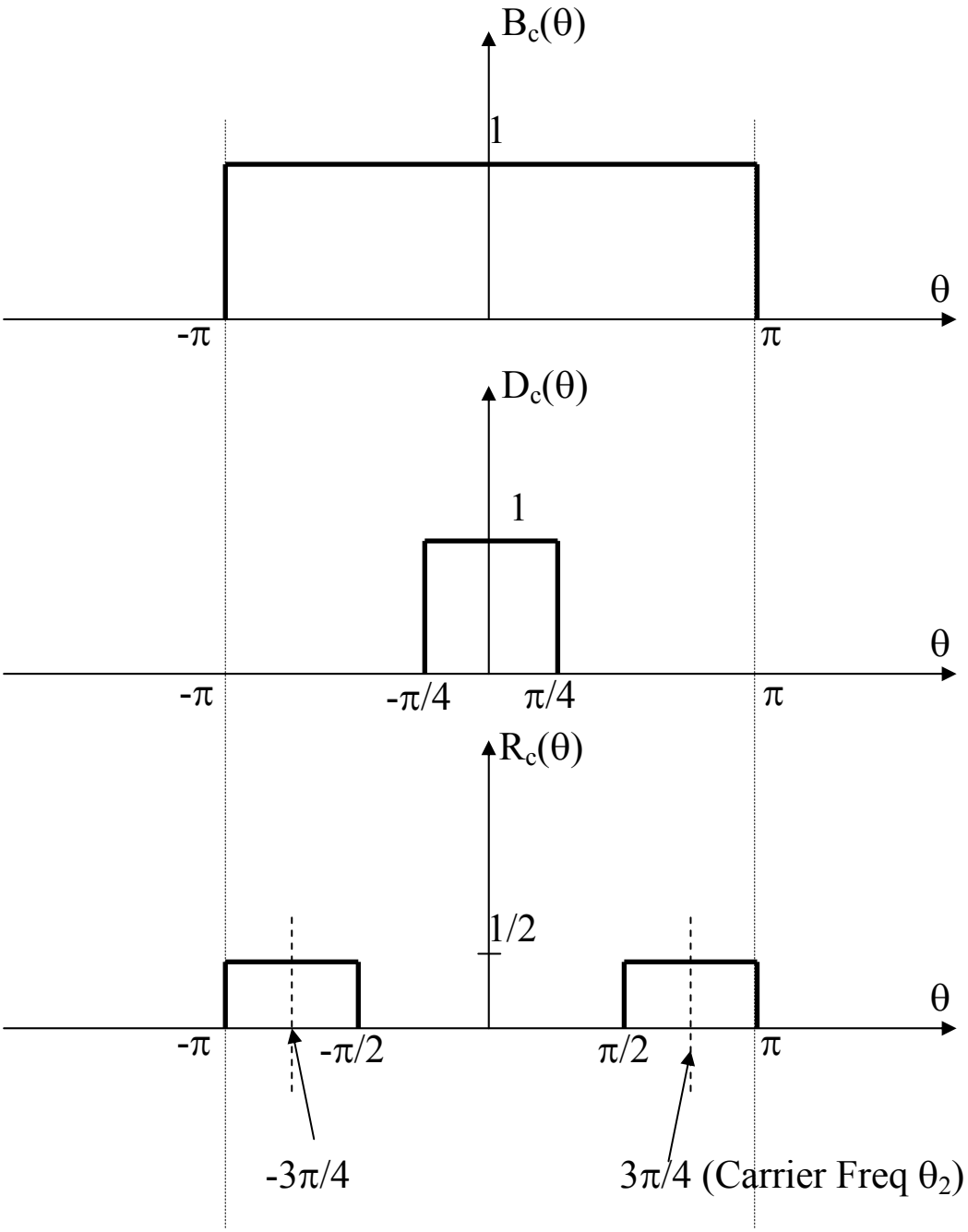


$$R_c(\theta) = \frac{1}{2}\{D_c(\theta - \theta_1) + D_c(\theta + \theta_1)\}$$

$$= \frac{1}{2}\left\{D_C(\theta - \frac{3\pi}{4}) + D_c(\theta + \frac{3\pi}{4})\right\}$$

$$= \begin{cases} \frac{1}{2} & \frac{-\pi}{4} \le (\theta - \frac{3\pi}{4}) \le \frac{\pi}{4} \\ \frac{1}{2} & \frac{-\pi}{4} \le \theta + \frac{3\pi}{4} \le \frac{\pi}{4} \\ 0 & \text{elsewhere} \end{cases}$$

$$= \begin{cases} \frac{1}{2} & \frac{\pi}{2} \le \theta \le \pi \\ \frac{1}{2} & -\pi \le \theta \le \frac{-\pi}{2} \\ 0 & \text{elsewhere} \end{cases}$$


$B_c(\theta)$
1
θ
-π
π
$D_c(\theta)$
1
θ
-π
-π/4
π/4
π
$R_c(\theta)$
1/2
θ
-π
-π/2
π/2
π
-3π/4
3π/4 (Carrier Freq θ_2)

(B)

$$P(z) = X(z)z^{-1} \Rightarrow P\left(z^{\frac{1}{2}}\right) = X\left(z^{\frac{1}{2}}\right) \cdot z^{-\frac{1}{2}}$$

$$\therefore P\left(-z^{\frac{1}{2}}\right) = X\left(-z^{\frac{1}{2}}\right) \cdot (-1)z^{-\frac{1}{2}}$$

$$W(z) = \frac{1}{2}\left[P\left(z^{\frac{1}{2}}\right) + P\left(-z^{\frac{1}{2}}\right)\right] = \frac{1}{2}\left[X\left(z^{\frac{1}{2}}\right) \cdot z^{-\frac{1}{2}} - X\left(-z^{\frac{1}{2}}\right) \cdot z^{-\frac{1}{2}}\right]$$

$$V(z) = \frac{1}{2}\left[X\left(z^{\frac{1}{2}}\right) + X\left(-z^{\frac{1}{2}}\right)\right]$$

$$Q(z) = V\left(z^2\right) = \frac{1}{2}\left[X(z) + X(-z)\right]$$

$$R(z) = W\left(z^2\right) = \frac{1}{2}\left[X(z) \cdot z^{-1} - X(-z) \cdot z^{-1}\right]$$

$$Y(z) = U(z) + R(z) = z^{-1}Q(z) + R(z)$$

$$Y(z) = \frac{z^{-1}}{2}\left[X(z) + X(-z)\right] + \frac{z^{-1}}{2}\left[X(z) - X(-z)\right] = X(z)z^{-1}$$

$$\therefore y(n) = x(n-1)$$